

Evaluating Identity-Based Bias in Toxicity Classification Models

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Abstract

Hate speech detection systems are commonly used on social media platforms to automatically flag harmful content. However, research has shown that these systems may unfairly label language associated with certain communities or dialects as toxic, even when it is not intended to be offensive. This project will examine whether hate speech classifiers produce higher false positive rates for identity-related terms or dialectal language. I plan to train supervised text classification models on standard hate speech datasets and evaluate their performance using accuracy and F1-score. In addition to overall performance, I will analyze subgroup metrics such as false positive rate differences across identity groups. Through subgroup analysis and counterfactual testing, this project aims to better understand how model design and dataset composition may contribute to biased outcomes in automated content moderation systems.

1 Introduction

The motivation for this project is to investigate fairness issues in hate speech classifiers and analyze how language variety, dialect, and community-specific terms influence model predictions. By examining bias patterns, this project aims to better understand the ethical tradeoffs between content moderation and fairness.

Primary Ethical Question: Do hate speech detection models unfairly classify language associated with specific communities or dialects as toxic or hateful?

2 Related Work

Prior research has shown that toxicity detection systems can learn unintended associations between identity terms and toxic language. The Jigsaw Unintended Bias in Toxicity Classification competition was motivated by the observation that toxicity models sometimes assign high toxicity scores

to non-toxic comments containing identity terms, such as comments mentioning LGBTQ identities or religious groups (Jigsaw, 2019). The Civil Comments dataset provides identity annotations that make it possible to evaluate subgroup performance and unintended bias in toxicity classification (Google, 2024).

Counterfactual fairness has also been studied in text classification. Garg et al. define counterfactual fairness in this setting as the question of whether a prediction would change if the sensitive identity term in the text were changed while the rest of the sentence remained similar (Garg et al., 2019). This idea motivates the counterfactual identity-swap test used in this project. More recent work has explored identity-based counterfactual data augmentation as a way to improve fairness by generating alternative versions of training examples with swapped identity terms (Wadhwa et al., 2022).

Recent surveys on bias and fairness in language models emphasize that bias can emerge from training data, model architecture, evaluation design, and deployment context (Gallegos et al., 2024). This is important for toxicity detection because high overall classification performance does not necessarily imply fair subgroup performance. Other studies have also shown that annotation practices and rater identity can affect toxicity labels, suggesting that toxicity datasets themselves may encode social and demographic biases (Goyal et al., 2022). This project builds on these concerns by comparing multiple classifiers and evaluating whether subgroup false positive disparities persist across models.

3 Methodology

The goal of this project is to determine whether toxicity classifiers produce unequal false positive rates for comments mentioning different identity groups. The methodology consists of four main components: model comparison, subgroup false

positive analysis, threshold sensitivity analysis, and counterfactual identity-swap testing.

First, multiple classifiers are trained to determine whether observed bias patterns are specific to one model or consistent across modeling approaches. Second, subgroup false positive rate is computed for each identity group. False positive rate is defined as:

$$FPR_g = \frac{FP_g}{FP_g + TN_g}$$

where FP_g is the number of non-toxic comments mentioning group g that are incorrectly predicted as toxic, and TN_g is the number of non-toxic comments mentioning group g that are correctly predicted as non-toxic. FPR disparity is computed as:

$$Disparity_g = FPR_g - FPR_{overall}$$

Third, threshold sensitivity analysis is used to evaluate how decision threshold affects both toxic recall and subgroup false positive rates. Finally, counterfactual identity-swap testing is used to measure whether model predictions change when identity terms are replaced with paired alternatives. The counterfactual flip rate is defined as the fraction of examples whose predicted label changes after identity-term swapping.

4 Experimental Setup

4.1 Dataset

Experiments were conducted using the Jigsaw Unintended Bias in Toxicity Classification dataset. Each comment includes a toxicity score and identity annotations for demographic groups. The comment text was renamed as `text`, and binary toxicity labels were created by assigning label 1 when the toxicity score was greater than 0.5 and label 0 otherwise. Identity columns were filled with 0 when missing. The dataset was split into 80% training and 20% testing using a fixed random seed.

Table 1: Training set class distribution

Class	Count	Percentage
Non-toxic (0)	1,358,633	94.1%
Toxic (1)	85,263	5.9%
Total	1,443,896	100.0%

The class distribution shows that the dataset is highly imbalanced, making accuracy alone insufficient for evaluating model behavior.

4.2 Models

Three traditional machine learning models were trained using TF-IDF features with a maximum vocabulary size of 10,000 and English stop-word removal. The models were Logistic Regression, Linear SVM, and Multinomial Naive Bayes. Logistic Regression and SVM used class balancing to account for class imbalance. The SVM model was calibrated to produce probability scores for threshold-based analysis.

4.3 Evaluation Metrics

Models were evaluated using accuracy, precision, recall, and F1-score. Because the research question focuses on fairness, additional metrics were computed: subgroup false positive rate, FPR disparity, threshold sensitivity, counterfactual flip rate, and identity-term ablation effects.

5 Results

5.1 Overall Classification Performance

Table 2: Overall model performance on the test set

Model	Acc.	Toxic Prec.	Toxic Rec.	Toxic F1
Logistic Reg.	0.830	0.24	0.91	0.38
SVM	0.957	0.65	0.58	0.61
Naive Bayes	0.951	0.85	0.20	0.32

Logistic Regression achieved the highest toxic recall, identifying 91% of toxic comments, but had very low toxic precision. This indicates that the model aggressively predicted toxicity and produced many false positives. SVM produced the strongest balance between precision and recall, while Naive Bayes achieved high precision but very low recall, meaning it missed most toxic comments.

Table 3: Detailed classification report by model

Model	Class	Prec.	Rec.	F1	Support
Logistic Reg.	Non-toxic	0.99	0.83	0.90	339,800
Logistic Reg.	Toxic	0.24	0.91	0.38	21,175
SVM	Non-toxic	0.97	0.98	0.98	339,800
SVM	Toxic	0.65	0.58	0.61	21,175
Naive Bayes	Non-toxic	0.95	1.00	0.97	339,800
Naive Bayes	Toxic	0.85	0.20	0.32	21,175

Table 4: Selected subgroup false positive rates by model

Model	Overall	Black	White	Muslim	Jewish	H/G/L
Logistic Reg.	0.175	0.718	0.676	0.622	0.480	0.676
SVM	0.019	0.149	0.107	0.084	0.059	0.137
Naive Bayes	0.002	0.018	0.022	0.008	0.007	0.010

5.2 Subgroup False Positive Rate

Subgroup FPR analysis shows that high overall performance can hide large disparities. Logistic Regression had an overall FPR of 0.175, but several identity groups had much higher FPRs. For example, non-toxic comments mentioning Black identity had an FPR of 0.718, and comments mentioning Muslim identity had an FPR of 0.622. SVM reduced the overall FPR substantially, but the same groups still had elevated false positive rates compared with the overall baseline. Naive Bayes had the lowest FPRs, but this result must be interpreted together with its low toxic recall.

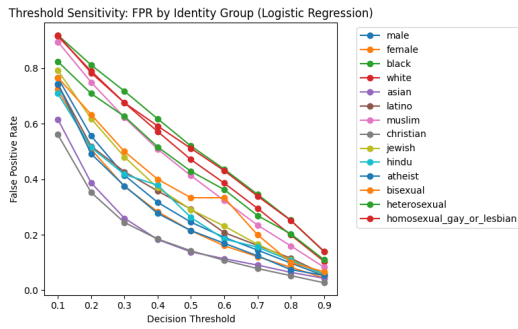


Figure 1: False positive rate by identity group for the Logistic Regression model. The dashed line represents the overall false positive rate.

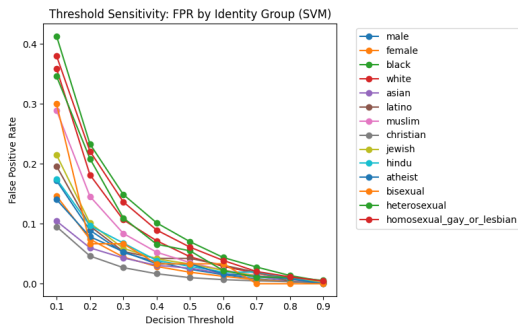


Figure 2: False positive rate by identity group for the SVM model. The dashed line represents the overall false positive rate.

5.3 Threshold Sensitivity

Threshold sensitivity analysis was conducted by varying the decision threshold from 0.1 to 0.9.

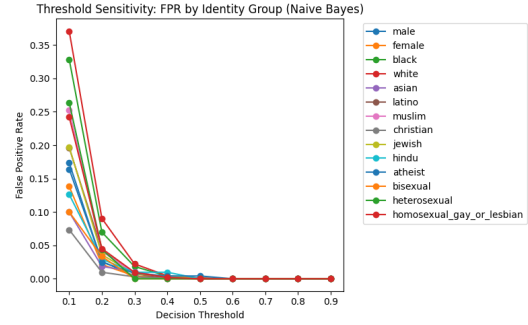


Figure 3: False positive rate by identity group for the Naive Bayes model. The dashed line represents the overall false positive rate.

Lower thresholds increased toxic recall but also increased false positive rates. For Logistic Regression at threshold 0.1, recall reached 0.980, but the overall FPR increased to 0.504 and toxic precision dropped to 0.108. This shows that aggressive thresholding can catch more toxic comments but creates substantial over-flagging.

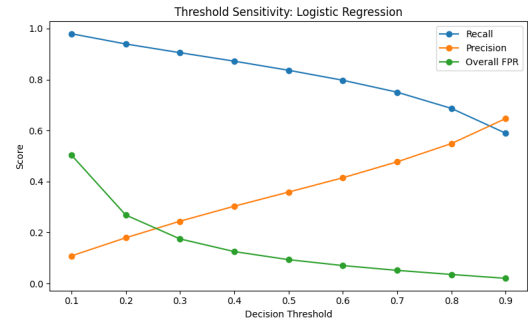


Figure 4: Threshold sensitivity showing the relationship between decision threshold, recall, and subgroup false positive rate.

5.4 Counterfactual Identity-Swap Results

Table 5: Counterfactual flip rates for selected identity groups

Model	Overall	Muslim	Black	Hetero.	H/G/L
Logistic Reg.	0.009	0.173	0.068	0.211	0.337
SVM	0.003	0.062	0.043	0.127	0.146
Naive Bayes	0.000	0.008	0.009	0.023	0.014

Although the overall flip rate was low for all models, flip rates were much higher for identity-specific subsets. Logistic Regression was especially sensitive to sexual orientation and religion swaps, with a flip rate of 0.337 for comments mentioning homosexual, gay, or lesbian identity and 0.173 for comments mentioning Muslim identity.

SVM reduced but did not eliminate this instability. Naive Bayes showed minimal flipping, likely because it rarely predicted comments as toxic.

6 Additional Analysis

6.1 Feature Importance

For Logistic Regression, TF-IDF feature coefficients were examined to identify the top features most predictive of toxicity. None of the top 50 positive toxicity features were direct identity terms.

Table 6: Identity-term representation among top toxic TF-IDF features

Feature Set	Identity Terms	Percent
Top 50 toxic features	0	0.0%

This suggests that identity terms were not the strongest global toxicity indicators. However, this does not mean that identity terms had no effect on subgroup predictions. A feature can have a moderate coefficient but still push borderline examples across the decision threshold.

7 Discussion

The results suggest that subgroup bias is both dataset-driven and model-dependent. Several identity groups, including Black, White, Muslim, heterosexual, and homosexual/gay/lesbian, had elevated false positive rates across multiple models. This consistency suggests that the dataset may contain patterns that lead classifiers to associate identity-related language with toxicity. However, the magnitude of the disparity varied significantly by model. Logistic Regression amplified false positive disparities, SVM reduced them, and Naive Bayes produced low FPRs largely because it rarely predicted the toxic class. Therefore, model choice and threshold choice both play important roles in fairness outcomes.

The feature importance and ablation results provide a more nuanced picture. Identity terms did not appear among the top 50 toxicity features, but removing identity terms reduced subgroup FPR across all groups. This suggests that identity terms may not be the dominant global toxicity features, but they still affect classification decisions for identity-related comments, especially near the decision threshold.

8 Conclusion

This project examined whether toxicity classifiers disproportionately flag non-toxic comments that mention protected identity groups. Results show that overall accuracy can hide substantial subgroup disparities. Logistic Regression achieved high toxic recall but produced large false positive disparities for identity-related comments. SVM improved the balance between precision and recall and reduced overall FPR, but it did not eliminate subgroup disparities. Naive Bayes produced the lowest false positive rates, but only by missing most toxic comments.

Counterfactual identity-swap testing showed that predictions were especially unstable for some religion and sexual orientation groups, while identity-term ablation showed that removing identity vocabulary reduced subgroup FPR across all groups. These findings suggest that identity-related language contributes to false positive disparities, but that bias is not caused only by direct identity terms. Future work should include transformer-based models such as DistilBERT, larger counterfactual swap dictionaries, confidence intervals for subgroup metrics, and mitigation techniques such as counterfactual data augmentation.

Acknowledgments

The code and intermediate results are available at: https://github.com/GustavoPCS/CS59300_Hate_Speech_Detection_Project.

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